

CS 3005

Theory of Automata

Lecture 23

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Turing Machine (TM)

The mathematical models (FAs, TGs, PDAs) that have been discussed so far can decide whether a string is accepted or not by them *i.e.* these models are language identifiers. However, there are still some languages which can't be accepted by them *e.g.* there does not exist any FA or TG or PDA accepting any non-CFLs.

Alan Mathison Turing developed the machines called **Turing Machine**, which accept some non-CFLs as well, in addition to CFLs.

Turing Machine

Definition: A Turing Machine (TM) consists of the following

1. An alphabet Σ of input letters.
2. An input TAPE partitioned into cells, having infinite many locations in one direction. The input string is placed on the TAPE starting its first letter on the cell i , the rest of the TAPE is initially filled with blanks (Δ 's).

Turing machine continued ...

Input TAPE

i	ii	iii	iv	
a	b	a	Δ	...



TAPE Head

3. A tape Head can read the contents of cell on the TAPE in one step. It can replace the character at any cell and can reposition itself to the next cell to the right or to the left of that it has just read.

Turing machine continued ...

Initially the TAPE Head is at the cell i . The TAPE Head can't move to the left of cell i . the location of the TAPE Head is denoted by .

4. An alphabet Γ of characters that can be printed on the TAPE by the TAPE Head. Γ may include the letters of Σ . Even the TAPE Head can print blank Δ , which means to erase some character from the TAPE.

Turing machine continued ...

5. Finite set of states containing exactly one START state and some (may be none) HALT states that cause execution to terminate when the HALT states are entered.
6. A **program** which is the set of rules, which show that which state is to be entered when a letter is read from the TAPE and what character is to be printed. This program is shown by the states connected by directed edges labeled by triplet
(letter, letter, direction)

Turing machine continued ...

It may be noted that the first letter is the character the TAPE Head reads from the cell to which it is pointing. The second letter is what the TAPE Head prints the cell before it leaves. The direction tells the TAPE Head whether to move one cell to the right, R, or one cell to the left, L. **Following is a note**

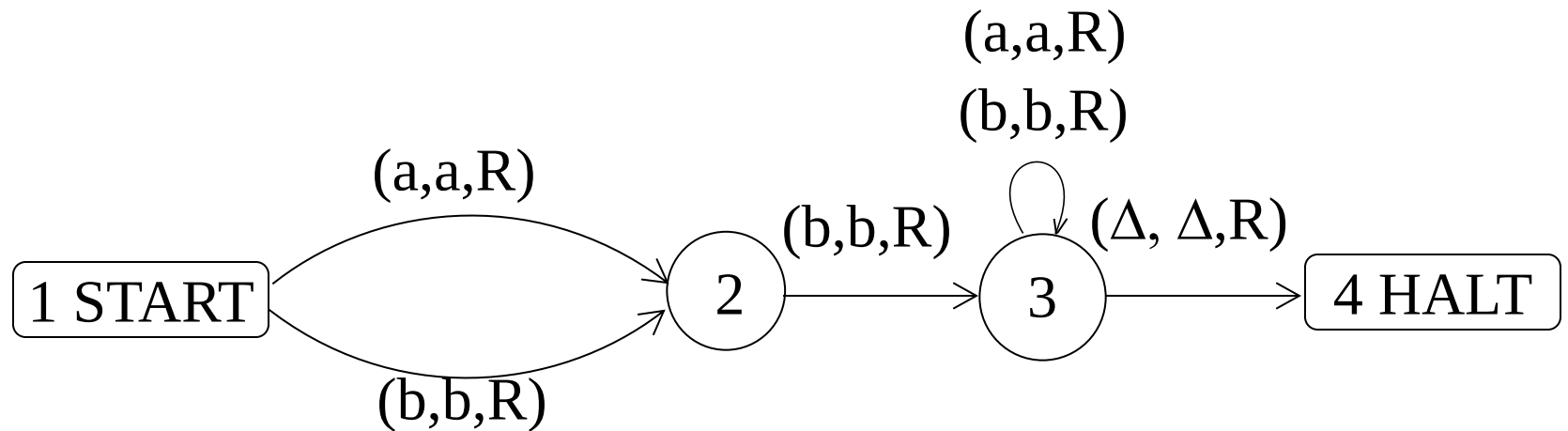
Note

It may be noted that there may not be any outgoing edge at certain state for certain letter to be read from the TAPE, which creates nondeterminism in Turing machines. It may also be noted that at certain state, there can't be more than one outgoing edges for certain letter to be read from the TAPE. The machine crashes if there is no path for a letter to be read from the TAPE and the corresponding string is supposed to be rejected.

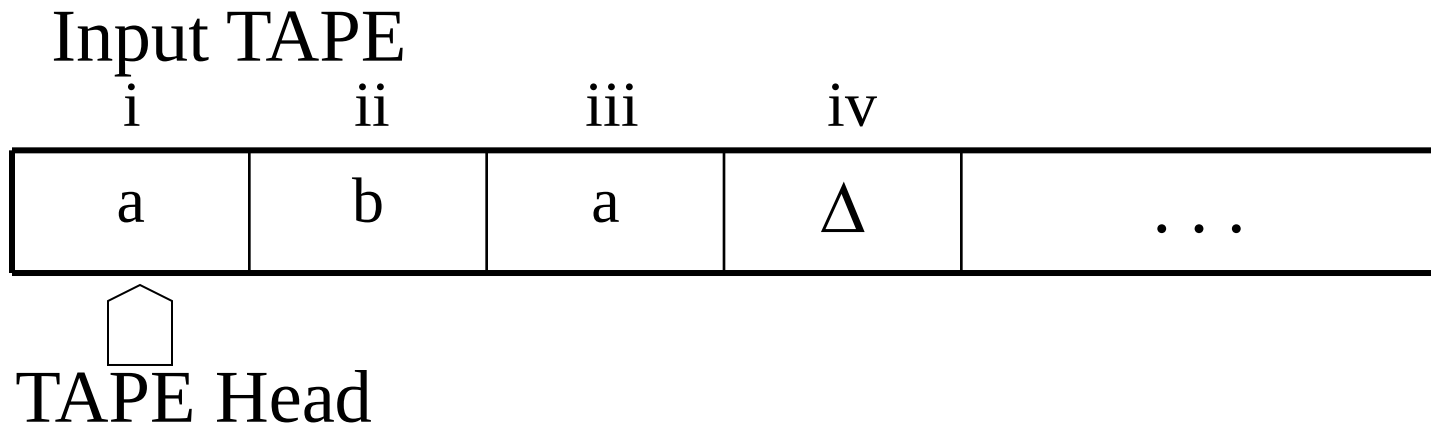
Example

Consider the following Turing machine

Let the input string aba be run over this TM



Example continued ...



Starting from the START state, reading a from the TAPE and according to the TM program, a will be printed *i.e.* a will be replaced by a and the TAPE Head will be moved one cell to the right.

Which can be seen as

Input TAPE

i	ii	iii	iv	
a	b	a	Δ	...



TAPE Head

This process can be expressed as

1		2
<u>a</u> ba	[?]	a <u>b</u> a

At state 2 reading b, state 3 is entered and the letter b is replaced by b, *i.e.*

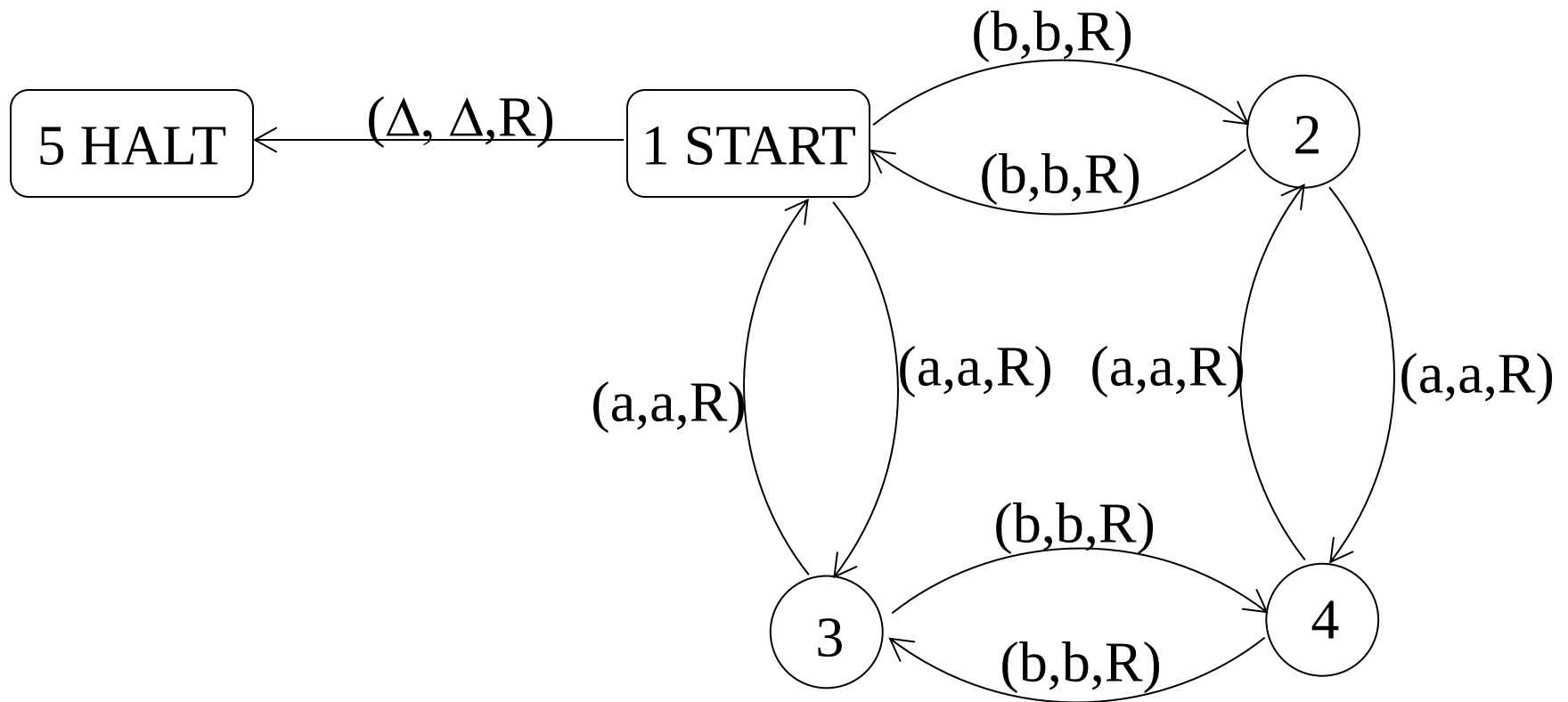
At state 3 reading a, will keep the state of the TM unchanged. Lastly, the blank Δ is read and Δ is replaced by Δ and the HALT state is entered. Which can be expressed as

1 2 3 3 HALT
 aba aba aba aba Δ

Which shows that the string aba is accepted by this machine. It can be observed, from the program of the TM, that the machine accepts the language expressed by $(a+b)b(a+b)^*$.

Theorem: Every regular language is accepted by some TM.

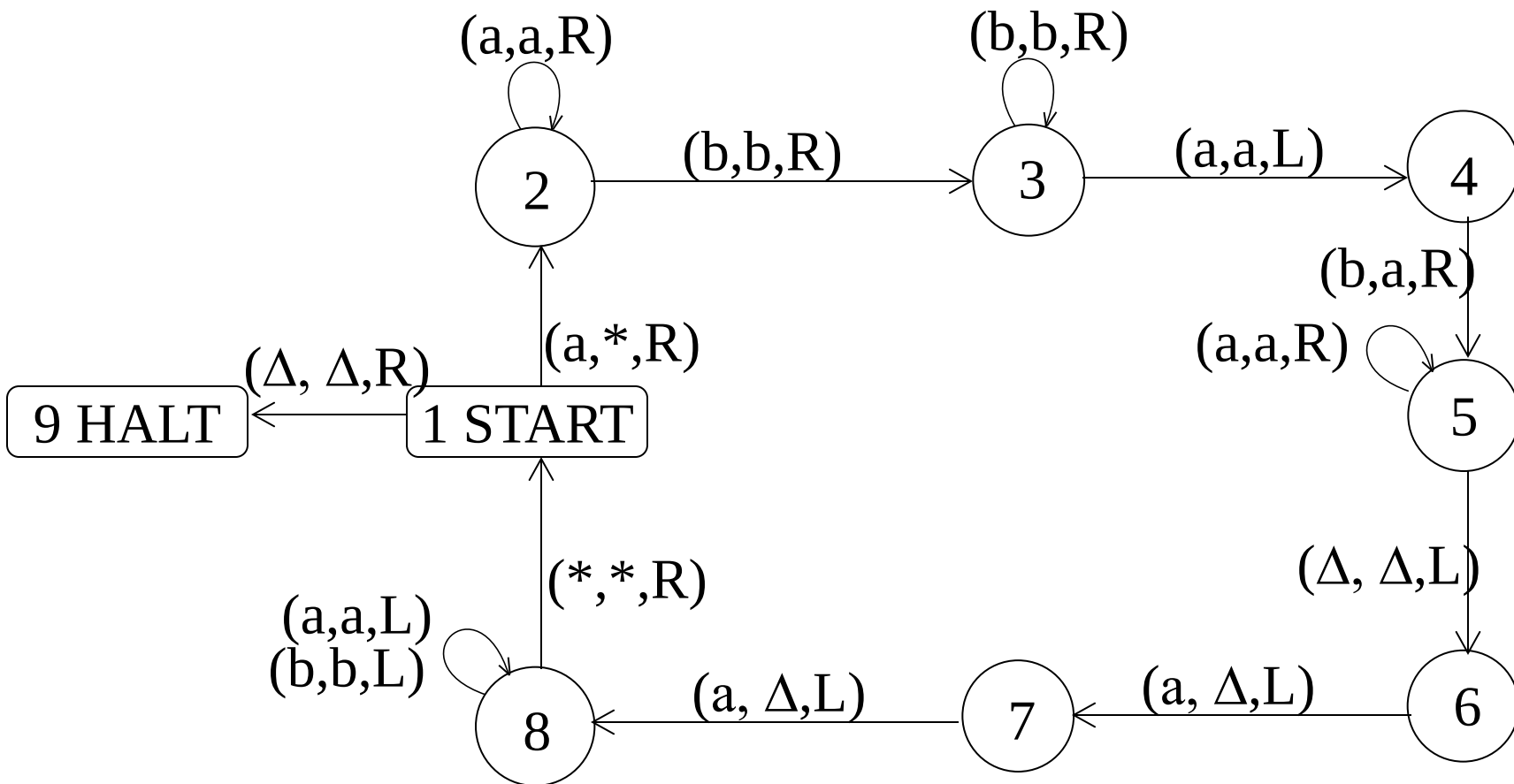
Example: Consider the EVEN-EVEN language. Following is a TM accepting the EVEN-EVEN language.



It may be noted that the above diagram is similar to that of FA corresponding to EVEN-EVEN language. Following is another example

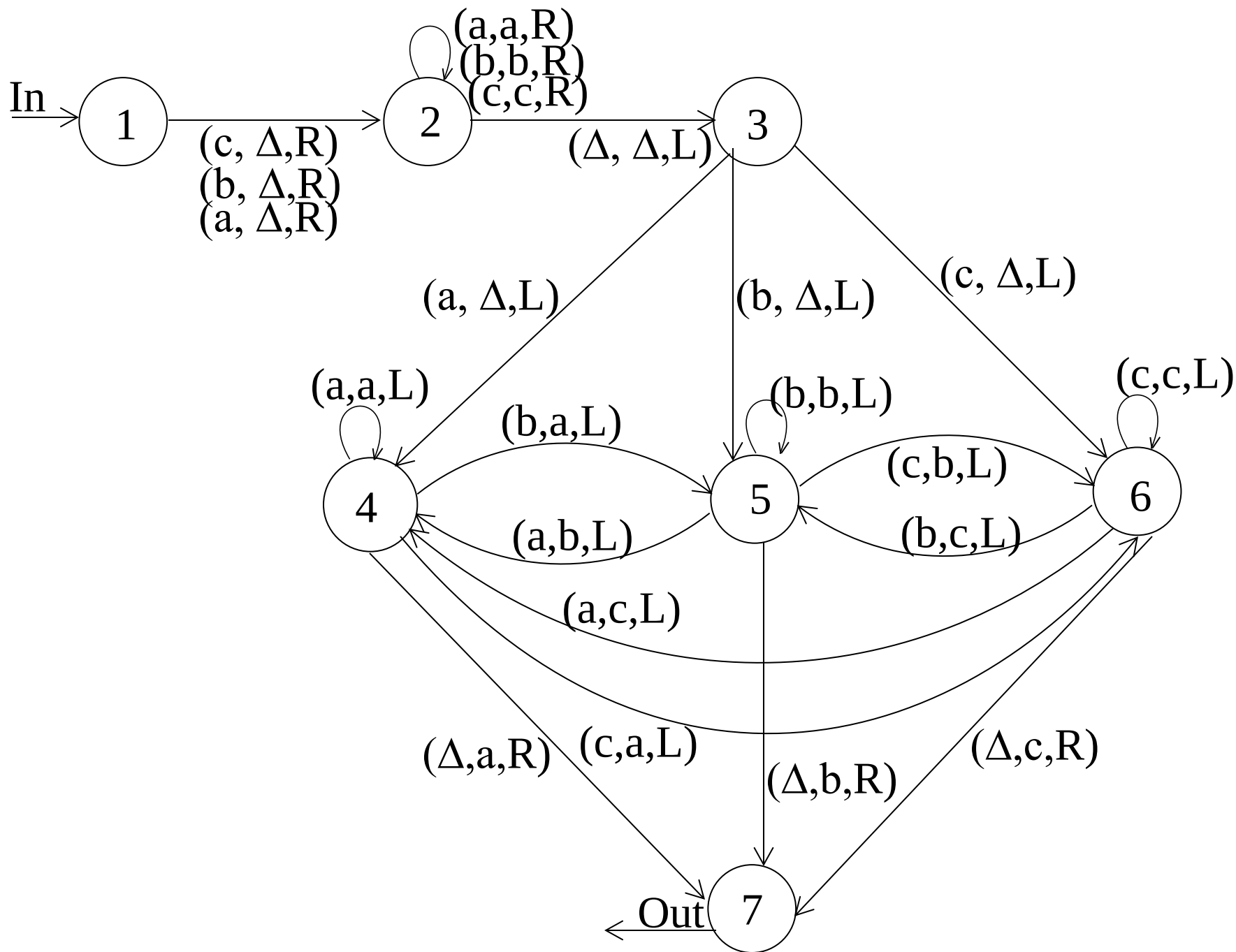
Example

Consider the following TM

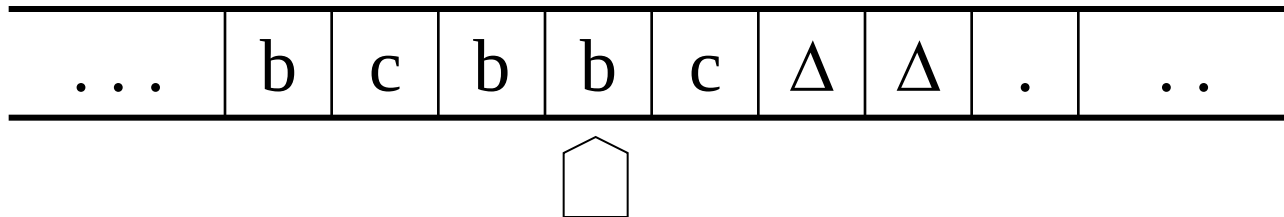


Example continued ...

The string aaabbbbaaa can be observed to be accepted by the above TM. It can also be observed that the above TM accepts the non-CFL $\{a^n b^n a^n\}$.



The process of deletion of letter a from the string bcabbc can easily be checked, giving the TAPE situation as shown below



Summing Up

Turing machine, examples, DELETE subprogram, example, INSERT subprogram, example.

A TM *recognizes* a language iff it accepts all and only those strings in the language.

language L is called **Turing-recognizable**
recursively enumerable
iff some TM recognizes L.

A TM *decides* a language L iff it **accepts** all strings in L and **rejects** all strings not in L.

A language L is called **decidable** or **recursive** iff some TM **decides** L.

A language is called Turing-recognizable or recursively **enumerable (r.e.)** if some TM recognizes it.

A language is called decidable or **recursive** if some TM decides it.

